English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, Software Engineering, Year 3, Groups 4/5, 2020/2021 Test 2

Your grade is 2 plus the number of points earned. Time allowed: 4 hours. Good luck.
Throughout, we work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Problem 1. (2 points)

1. (0.5 points) Two cards are drawn from a standard 52-card deck. Let X be the number of aces and Y the number of hearts. Find the joint distribution and correlation of X, Y.

   • Let $C = \binom{52}{2} = 1326$. Combinatorial counting gives

X (aces)\Y (hearts)	0	1	2
0	$\binom{36}{2}/C$	$\binom{12}{1}\binom{36}{1}/C$	$\binom{12}{2}/C$
1	$\binom{3}{1}\binom{36}{1}/C$	$(\binom{12}{1}\binom{3}{1}+\binom{1}{1}\binom{36}{1})/C$	$\binom{12}{1}/C$
2	$\binom{3}{2}/C$	$\binom{3}{1}/C$	0

   ,

   After calculating and summing the rows and columns (or directly by counting),

X (aces)\Y (hearts)	0	1	2
0	105/221	72/221	11/221
1	18/221	12/221	2/221
2	1/442	1/442	0

Y (hearts)	0	1	2
$\mathbb{P}(Y=\cdot)$	19/34	13/34	2/34

X (aces)	0	1	2
$\mathbb{P}(X=\cdot)$	188/221	32/221	1/221

   .

   Therefore

$$\mathbb{E}X = 34/221 = 2/13; \qquad DX = \mathbb{E}X^2 - (\mathbb{E}X)^2 = 36/221 - (2/13)^2 = 400/2873;$$
$$\mathbb{E}Y = 17/34 = 1/2; \qquad DY = \mathbb{E}Y^2 - (\mathbb{E}Y)^2 = 21/34 - 1/4 = 25/68;$$
$$\mathbb{E}XY = 12/221 + 4/221 + 2/442 = 1/13; \qquad Cor(X,Y) = \frac{\mathbb{E}XY - \mathbb{E}X\mathbb{E}Y}{\sqrt{DX \cdot DY}} = \frac{1/13 - 1/13}{\sqrt{DX \cdot DY}} = 0\,.$$

   Thus X, Y are uncorrelated, although they are not independent!

2. (1 point) Let X be an asset price and Y the average deposit interest rate (both daily and in dimensionless units). Their joint density is $f_{X,Y}(x,y) = c(x^2 + e^y x)$ for $0 < x, y < 1$, and zero elsewhere, where c is a constant. Find c and the covariance of X, Y. What is the expected asset price if the interest rate is 0.5?

   •
$$1 = c\int_0^1\int_0^1 (x^2 + e^y x)\,\mathrm{d}x\,\mathrm{d}y = c\int_0^1 \frac{1}{3} + \frac{1}{2}e^y\,\mathrm{d}y = c\left(\frac{1}{3} + \frac{1}{2}(e-1)\right) = \frac{3e-1}{6}\,.$$

   Hence $c = 6/(3e-1)$. For $x \in (0,1)$,

$$f_X(x) = \int_0^1 c(x^2 + e^y x)\,\mathrm{d}y = c(x^2 + (e-1)x)\,.$$

   Therefore

$$\mathbb{E}X = c\int_0^1 x(x^2 + (e-1)x)\,\mathrm{d}x = c\left(\frac{1}{4} + (e-1)\frac{1}{3}\right) = c\frac{4e-1}{12}\,.$$

   Similarly, for $y \in (0,1)$,

$$f_Y(y) = c\int_0^1 x^2 + e^y x\,\mathrm{d}x = c\left(\frac{1}{3} + \frac{1}{2}e^y\right) = \frac{c}{6}\left(2 + 3e^y\right)\,,$$

$$\mathbb{E}Y = \frac{c}{6}\int_0^1 y(2 + 3e^y)\,\mathrm{d}y = \frac{c}{6}\left(2\frac{1}{2} + 3\right) = \frac{2c}{3}\,,$$

where we used $\int xe^x \,\mathrm{d}x = \int x \,\mathrm{d}e^x = xe^x - \int e^x \,\mathrm{d}x = e^x(x-1)$ and hence $\int_0^1 xe^x \,\mathrm{d}x = 1$. Next,

$$\mathbb{E}XY = c\int_0^1 \int_0^1 xy(x^2 + e^y x)\,\mathrm{d}x\,\mathrm{d}y = c\int_0^1 \frac{1}{4}y + \frac{1}{3}ye^y \,\mathrm{d}y = c\left(\frac{1}{4}\cdot\frac{1}{2} + \frac{1}{3}\right) = \frac{11c}{24}\,.$$

Therefore

$$Cov(X,Y) = \mathbb{E}XY - \mathbb{E}X\mathbb{E}Y = \frac{11c}{24} - c^2\frac{4e-1}{18} \approx -0.0014\,.$$

For the last question,

$$\begin{aligned}\mathbb{E}(X|Y=0.5) &= \int_0^1 x f_{X|Y}(x|y=0.5)\,\mathrm{d}x = \int_0^1 x\frac{f_{X,Y}(x,0.5)}{f_Y(0.5)}\,\mathrm{d}x = \int_0^1 x\frac{c(x^2+x\sqrt{e})}{\frac{c}{6}(2+3\sqrt{e})}\,\mathrm{d}x \\ &= \frac{6}{2+3\sqrt{e}}\int_0^1 x^3 + x^2\sqrt{e}\,\mathrm{d}x = \frac{6}{2+3\sqrt{e}}(\frac{1}{4} + \frac{1}{3}\sqrt{e}) \approx 0.6907\,.\end{aligned}$$

3. (0.5 points) Let X have density $f_X(x) = \frac{1}{4}xe^{-x/2}$ for $x \geq 0$ and zero otherwise. Find the density of $Y = -2X + 2$.

   • Apply the transformation theorem directly or follow its proof: for $y > 2$, $\mathbb{P}(Y \leq 2) = \mathbb{P}(X \geq 0) = 1$ and $f_Y(y) = 0$.

   For $y \leq 2$, $F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(-2X + 2 \leq y) = \mathbb{P}(X \geq 1/2(2-y)) = 1 - F_X(1/2(2-y))$. Therefore

$$\begin{aligned}f_Y(y) = \frac{\mathrm{d}}{\mathrm{d}y}F_y(y) &= \frac{\mathrm{d}}{\mathrm{d}y}\left(1 - F_X\left(\frac{1}{2}(2-y)\right)\right) = -F_X'\left(\frac{1}{2}(2-y)\right)\cdot\left(\frac{1}{2}(2-y)\right)' \\ &= \frac{1}{2}f_X\left(1 - \frac{1}{2}y\right) = \frac{1}{2}\cdot\frac{1}{4}\left(1 - \frac{1}{2}y\right)e^{y/4 - 1/2}\,.\end{aligned}$$

Problem 2. (0.5 points)

According to official data, Sofia's historical average January temperature is $-0.5°C$. Let X be the average January temperature in Sofia in degrees Celsius, assumed normally distributed. The variance is usually estimated from data too, but here we consider a different (not recommended) method.

Suppose someone regards a temperature above $20°C$ as an extreme anomaly with probability 1 percent. What variance estimate does this imply? What is then the probability of a temperature below $-15°C$?

• Let σ^2, $\sigma \geq 0$, be the variance, so $X \sim N(-0.5, \sigma^2)$. We have

$$\frac{1}{100} = \mathbb{P}\left(X \geq 20\right) = \mathbb{P}\left(\frac{X + 0.5}{\sigma} \geq \frac{20.5}{\sigma}\right) = \mathbb{P}\left(N(0,1) \geq \frac{20.5}{\sigma}\right) = 1 - \Phi\left(\frac{20.5}{\sigma}\right)\,,$$

where $\Phi(x) = \mathbb{P}(N(0,1) \leq x)$. From a normal table, find z such that $\Phi(z) = 0.99$. This gives $20.5/\sigma \approx 2.33$ and $\sigma \approx 8.80$. Then

$$\mathbb{P}(X \leq -15) = \mathbb{P}\left(\frac{X + 0.5}{8.80} \leq \frac{-14.5}{8.80}\right) \approx \mathbb{P}\left(N(0,1) \leq -1.65\right) \approx 0.0495\,.$$

Problem 3. (1.5 points) A point A is chosen randomly in the disk $k(O,1)$ with centre O and radius 1. Let $X = |OA|$. Can you guess the mode and median? Justify your guesses. What would you expect $\mathbb{E}X$ to be? *(The mode of a discrete distribution is a value with maximum probability. In the continuous case, analogously, we seek a value maximizing f_X. We call a a median of X if $\mathbb{P}(X \leq a) = \mathbb{P}(X \geq a) = 1/2$.)*

• Informally, we may guess that the mode is 1, because at a fixed distance from the centre, a larger distance corresponds to more points. Formally, the cardinalities are of course the same. The median is the value satisfying $\pi r^2 = 1 - \pi r^2$.

1. (0.75 points) Find the distribution function, density, expectation and variance of X.

   • For $c \in (0,1)$, $F_X(c) = \mathbb{P}(X \leq c) = \mathbb{P}(X \in k(O,c)) = S_{k(O,c)}/S_{k(0,1)} = c^2$. Hence $f_X(c) = F_X'(c) = 2c$. For completeness, if $c \geq 1$, then $F_X(c) = 1, f_X(c) = 0$, while if $c \leq 0$, $F_X(c) = 0$, $f_X(c) = 0$. We obtain

$$\mathbb{E}X = \int_0^1 x2x\,\mathrm{d}x = \frac{2}{3}; DX = \int_0^1 x^2\cdot 2x\,\mathrm{d}x - \frac{4}{9} = \frac{1}{18}\,.$$

2. (0.75 points) Now let A_1, A_2, A_3 be three points chosen randomly and independently in the same disk. What is the expected distance to the point nearest the centre? And to the farthest? *(Bonus: Find the expected distance to the middle point. Should it equal $\mathbb{E}X$?)*

• Let the distances from O to A_1, A_2, A_3 be X_1, X_2, X_3, respectively. Then $X_1, X_2, X_3 \sim X$ are independent, where X is as in the previous part. The distance to the nearest point is $Y_1 = \min(X_1, X_2, X_3)$, and to the farthest, $Y_2 = \max(X_1, X_2, X_3)$. For $y \in (0, 1)$, $\mathbb{P}(Y_2 \leq y) = \mathbb{P}(X \leq y)^3 = y^6$. Therefore

$$f_{Y_2}(y) = 6y^5; \mathbb{E}Y_2 = \int_0^1 y 6y^5 \, \mathrm{d}y = \frac{6}{7}.$$

Similarly, $\mathbb{P}(Y_1 \leq y) = 1 - \mathbb{P}(Y_1 \geq y) = 1 - (\mathbb{P}(X \geq y))^3 = 1 - (1 - y^2)^3$. Therefore

$$f_{Y_1}(y) = 6y(1 - y^2)^2; \mathbb{E}Y = \int_0^1 y 6y(1 - y^2)^2 \, \mathrm{d}y = \frac{16}{35}.$$